

Addendum: Economic Logic & Tiered Context Design

Extension to Flue Agent Framework Migration

1. The "Microscope to Telescope" Context Model

To ensure profitability on flat-rate subscriptions, the agent utilizes **Demand-Driven Context Fetching**. This prevents "Token Bloat" by only loading data as the reasoning logic requires it.

Tiered Data Access Levels

View	Scope	Access Trigger	Cost Profile
Microscope	Today's Recovery + Planned Session	Default (Every Wake-up)	Negligible (<1k tokens)
Magnifying Glass	Last 7 Days (Volume/ Fatigue)	Skill: <code>get_weekly_stats()</code>	Moderate (Llama-3.1-8b)
Telescope	Macrocycle Trends / Limiters	Skill: <code>query_long_term_trends()</code>	High (Vectorize / Sonnet)

2. Economic Guardrails & Model Selection

The system defaults to a "Compute-Efficient" model, escalating to high-reasoning models only when validated by a controller.

- **Default Model:** Claude 3.5 Haiku or Llama 3.1 70B (via Workers AI).
- **The "Switch" Logic:** If the user request involves complex biomechanical analysis or significant program pivots, the agent triggers a Flue Skill to re-initialize the session with **Claude 3.5 Sonnet**.
- **Usage Quotas:** D1-backed tracking of "Token Credits" per athlete per billing cycle.

3. "Watching the Watchers": The Supervisor Loop

An asynchronous Quality Assurance (QA) loop ensures that low-cost model interactions maintain the "Coach" standard.

```
// Scheduled Cron Trigger: Weekly Audit export default { async scheduled(event, env) {
const auditSample = await env.DB.prepare( "SELECT transcript FROM CoachInteractions WHERE
status = 'completed' LIMIT 10" ).all(); // High-reasoning model reviews the cheap model's
performance const report = await auditAgent.prompt("Analyze these interactions for safety,
tone, and logic accuracy.", { context: auditSample }); if (report.driftDetected) { await
env.BUCKET.put('users/global/directives_update.md', report.suggestedFixes); } } }
```

4. Summary of Benefits

- **Performance:** Faster Time-To-First-Token (TTFT) by minimizing initial context.
- **Scalability:** Prevents "Context Window Saturation" as athlete history grows over years.
- **Profitability:** Decouples user growth from API cost growth via tiered memory compression.